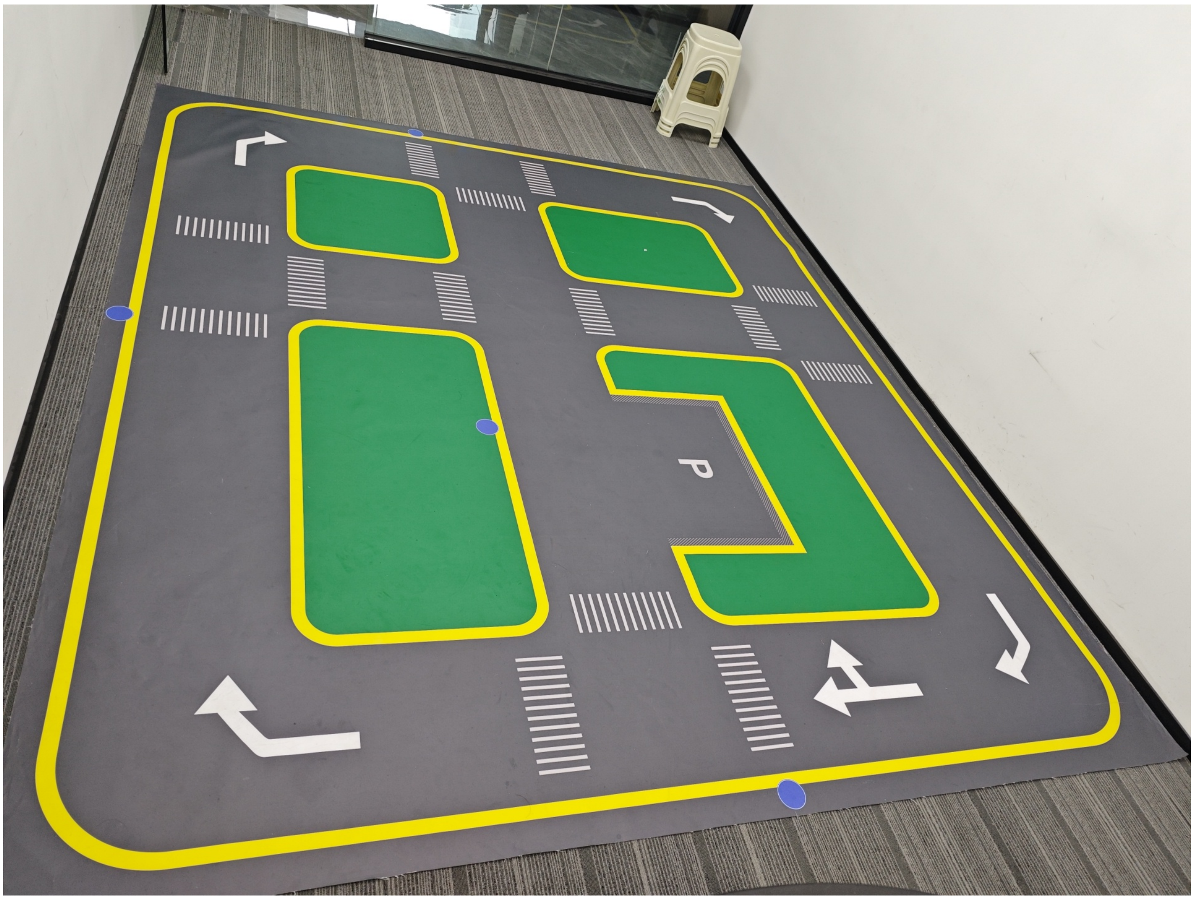
Site Layout and Camera/LiDAR Model Configuration

1. Site Layout

- Choose flat ground for map placement as much as possible. After unfolding, stretch the four corners of the map to prevent wrinkles.
    - If the map is not flat, lane detection may fail to detect lane lines on uneven parts during lane line detection.



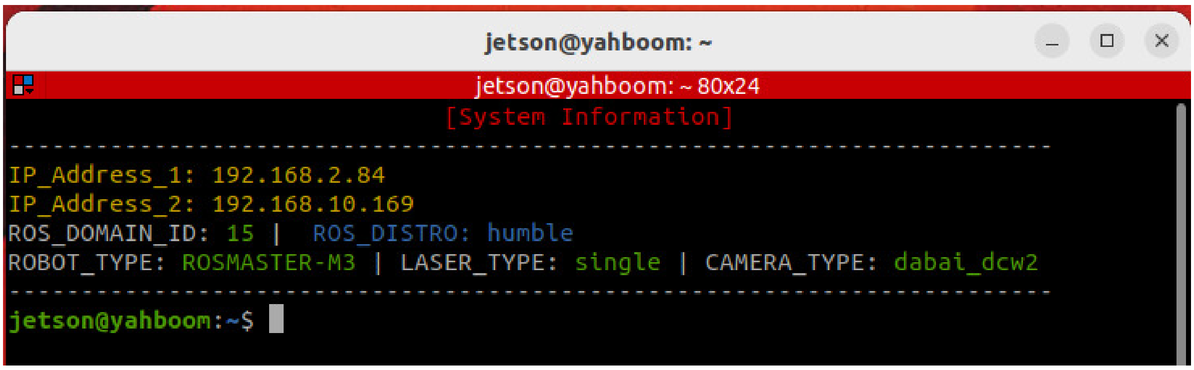
- Traffic sign placement: You can place traffic signs at the blue dot positions on the map according to your own needs. In this chapter's `11-Autonomous Driving Comprehensive Application` course, a reference autonomous driving scenario is provided.

> [!NOTE]
>
> - PA, PB parking signs, and right turn signs are implemented through specific action groups. If the placement positions differ from the diagram below, to achieve the best results, please refer to `7. Right Turn Sign Debugging` and `8. Parking Sign`

> `Debugging` in this chapter's `12-Autonomous Driving Debugging Special Topics` course. Optimize parameters through the visual `rqt_configure` panel.

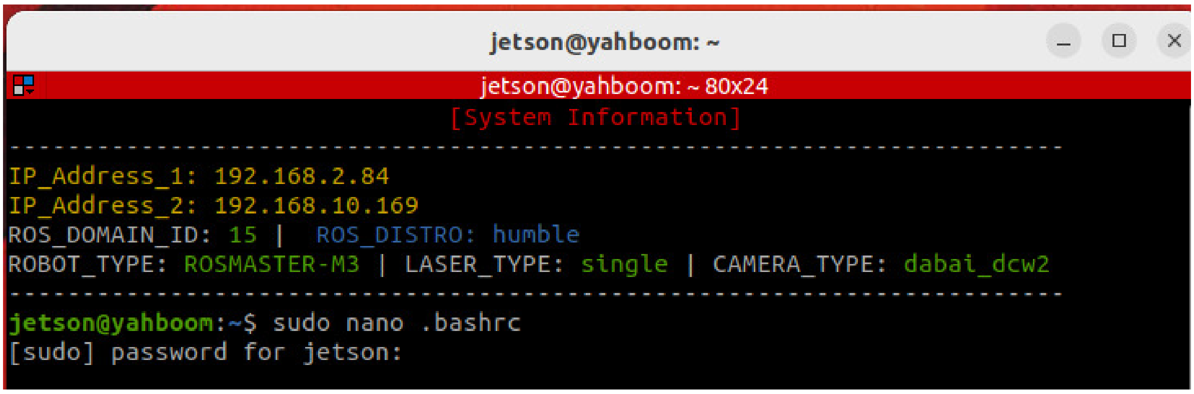
2. Setting Camera Model

- If you haven't set the camera model yet, you need to set your own camera and LiDAR models first. For example, in the diagram below, opening a new terminal will print system-related information.
  - LASER_TYPE: LiDAR model, distinguishing between single LiDAR version and dual LiDAR version
  - CAMERA_TYPE: Camera model, distinguishing between Angstrong Nuwa-HP60C and Orbbec DaBai DCW2

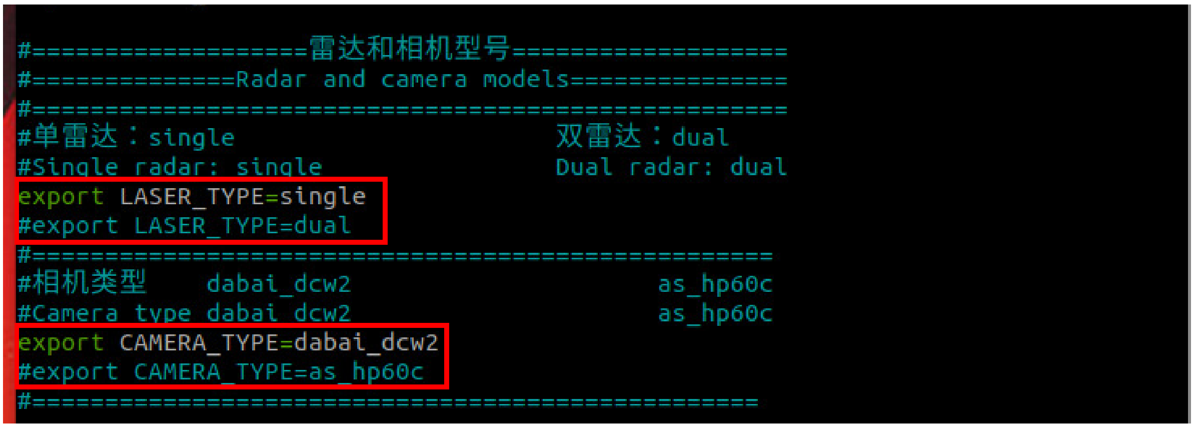


- Open the `.bashrc` file to set environment variables. The user password is yahboom.

```
sudo nano .bashrc
```



Scroll down to find the LiDAR and camera models. Comment or uncomment the corresponding environment variables according to your camera and LiDAR models.



- Press ctrl+x, then enter Y to save the file. Enter the following command to refresh the environment variables. This completes the hardware sensor configuration.

```
source .bashrc
```